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Published: 11 August 2022

ICCA 2022: 2nd International Conference on Computing Advancements
March 10 - 12, 2022
Dhaka, Bangladesh

ICCA '22: Proceedings of the 2nd International Conference on Computing Advancements (March 2022)
https://doi.org/10.1145/3542954.3543024
ISBN: 9781450397346

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ABSTRACT

Nowadays, class imbalance problem is one of the most important affairs among machine learning and data mining researchers. In this problem, majority of the sample data are labeled as one class and only a few samples are labeled as another class. This creates an imbalance in the dataset, and it is hard to solve because common classification algorithms are not designed to face these sorts of data. It means, the machine shows a result that is more influenced by the majority class, while the minority class is neglected. In this review paper we explore the main issues that come with the class imbalance problem, we also discuss different techniques to handle this problem. A fair knowledge about the multiclass imbalance problem is also given. In the end, we indicate some opportunities and challenges for future research in this domain.

CCS CONCEPTS

• **Artificial intelligence** → **Machine learning**; *Class imbalance problem.*

KEYWORDS

Class Imbalance Problem, SMOTE, Ensemble, Undersampling, Oversampling, Multiclass Imbalance

ACM Reference Format:

Nazim Uddin Niaz, K.M. Nadim Shahariar, and Muhammed J. A. Patwary. 2022. Class Imbalance Problems in Machine Learning: A Review of Methods And Future Challenges. In *2nd International Conference on Computing Advancements (ICCA 2022), March 10–12, 2022, Dhaka, Bangladesh.* ACM, New York, NY, USA, 6 pages. https://doi.org/10.1145/3542954.3543024

1 INTRODUCTION

In recent years, the world has seen the rapidly growing development of science and technology. This rapid growth has given an opportunity and availability of a massive amount of raw data. But this data is not always distributed equally. This unequal distribution creates a problem that is called the class imbalance problem [20]. This is a classification problem. There are mainly two or more classes, if there is two class then it is called binary class imbalance problem and if there is more than two class then it is called multiclass imbalance problem [39]. Classes are divided into majority class and minority class. The difference between labeled examples in the majority and minority classes is huge. Sometimes the ratio can be 1000:5 or even less. For this huge imbalance of data, traditional machine learning algorithms like to ignore the minority class. This type of problem can be found in real-world uses, like cancer detection, credit card fraud detection, stock market prediction, etc. The main issues that comes with the class imbalance problem:

Figure 1: Class Imbalance Problem

1. The dataset is highly imbalanced.
2. Because of that highly imbalanced dataset traditional algorithms don't work well.
3. Almost all the given solutions are for binary class imbalance and multiclass imbalance problem is being neglected.
4. There are no Standardized evaluation metrics.

Here, we seek to provide an appropriate survey over the solutions to the class imbalance problem and will also provide the benefits and drawbacks of these solutions for future scholars in the research of machine learning. Furthermore, we also provide a fair knowledge about the issue of multiclass imbalance and some future challenges for the class imbalance problem. Our contributions in this paper are:
1. To offer a detailed survey of techniques to work with the class imbalance problem.
2. To discuss the multiclass imbalance problem.
3. To give some future challenges for future research.
This paper is ordered as follows: Section 2 discusses the assessment metrics for the imbalance problem. Section 3 shows the techniques available for solving the class imbalance problem. Section 4 gives an overview of the multiclass imbalance problem. Section 5 shows some opportunities and future challenges for the class imbalance problem. Finally, Section 6 concludes the paper.

ICCA 2022, March 10–12, 2022, Dhaka, Bangladesh

ACM ISBN 978-1-4503-9734-6/22/03…$15.00
https://doi.org/10.1145/3542954.3543024

2 ASSESMENT METRICS

It is the most important parameter of a machine learning algorithm because by this we get to know how an algorithm is performing. Accuracy and error rate are the most familiar metrics used to evaluate an algorithm. But they do not perform great in the class imbalance problem. Because traditional classification algorithms work based on handling the equal distributions of data but in imbalance dataset data is highly skewed. for binary class, problem accuracy can be determined through the confusion matrix.

Table 1: Confusion Matrix

	Predictive Negative	Predictive Positive
Actual Negative	True Negative (TN)	False Positive (FP)
Actual Positive	False Negative (FN)	True Positive (TP)

As predictive accuracy performs poorly, researchers of the machine learning community came up with new metrics for imbalanced datasets from other domains like precision, recall, ROC, AUC (from medical domain), F-Measure, G-mean.

The ROC curve and the correlated AUC are also renowned metrics for classifiers. The ROC diagram is designed by arranging TP rate over FP rate. In highly imbalance dataset, ROC AUC curve performs better. Precision recall curve or PR-Curve is also can be used for class imbalance problem. It is the representation of precision (y-axis) and recall (x-axis). It is more appropriate to use than ROC curve in class imbalance problem. PR-Curve is useful when in the imbalance dataset negative samples are more than positive samples.

3 STRATEGIES FOR HANDLING IMBALANCED DATA

There are many proposed solutions to handle the class imbalance problem, all those solutions mainly fall under three types of approach to manage the class imbalance problem.

1. Data level: In data level methods, the first thing to do is remove the imbalance distribution of data. Because without balancing the data, standard classification algorithms will not work that great. Balancing can be done by generating new samples for the minority group (over-sampling) or by removing samples within the majority group (under-sampling). But this sometimes deletes the important samples or creates some meaningless new samples.
2. Algorithm level: This works by changing the algorithm, so to do this someone needs to have a better level of understanding in modifying algorithms. The most popular techniques of this algorithm level are: Cost sensitive approach, One class learning.
3. Hybrid: This is a combination of data level techniques and algorithmic level techniques working together for a better accurate model. Some hybrid techniques are: Easy ensemble, Balance cascade.

3.1 Data Level

Random undersampling and random oversampling are two primary techniques to work out the class imbalance problem. These techniques randomly delete or adds samples to balance the classes [6].

Table 2: Different Assessment Metrics

Name	Equation	Description
Accuracy	$\frac{(TP+FP)}{(TP+FP+TN+FN}$	Accuracy is defined as correct prediction rate for the test data.
Precision	$\frac{(TP)}{(TP+FP)}$	Positive predictive value or precision in the ration between true positives (TP) and all the positives (TP+FP)
Recall	$\frac{(TP)}{(TP+FN)}$	Sensitivity or TP Rate or recall is the measure of correctly identifying the true positives.
F-measure	$2 * \frac{(Precision*Recall)}{(Precision+Recall)}$	F-measure is a metric for determining how accurate a model is on a given dataset.
G-mean	$\sqrt{(\frac{(TP)}{(TP+FN)}) * (\frac{(TN)}{(TN+FP)})}$	Geometric Mean or G-mean is a metric that measures the balance within majority class and minority class.

Elimination of samples in random undersampling do not perform well because there is a waste of important information when samples are being eliminated in random order. It's also the same case for random oversampling, when increasing samples randomly in the minority class intakes more computational power than undersampling but performance is similar to undersampling [6]. These techniques are very time consuming, especially when working with large-scale datasets. Undersampling methods can be also solved by data cleaning techniques. The main target of these techniques is to identify the outlier, redundant and noisy patterns of the dataset and remove them for better performance of the classifier. Edited Nearest-Neighbor rule, Condensed Nearest Neighbour Rule (CNN), and Tomek-link are based on this technique. There is even a modified version of Tomek-link that is a Redundancy-driven modified Tomek-link [11]. There is another strategy that is the combination of Tomek-link and CNN rule named One-Sided Selection (OSS). Other techniques to solve class imbalance using undersampling is

Table 3: Advantages and Disadvantages of Different Level Methods

Method	Advantages	Disadvantages
Data Lavel	(1) Implementation is easy compared to other techniques. (2) Independent on the underlying classifier.	(1) May delete important samples and useful information in undersampling. (2) May lead to overfitting. (3) Extremely time-consuming. 4. Need more computational power.
Algorithmic Lavel	(1) Minimize the cost of misclassification with cost-sensitive learning. (2) User can set specific goals by modifying the algorithm. (3) One class learning performs better in highly imbalanced data.	(1) In cost-sensitive learning misclassification cost is usually unknown. (2) Needs better knowledge in learning algorithms to modify the algorithm.
Hybrid	(1) Performs better than many independent classifiers. (2) Can be used to solve complex problems.	(1) Can be time-consuming. (2) It is difficult to find the right amount of oversampling or undersampling to apply.

Inverse Random Undersampling (IRUS) [37], CSMOUTE [21].
In oversampling, Random oversampling techniques copy the instances of the minority class randomly which led to the overfitting problem to prevent this problem SMOTE [6] was proposed. In SMOTE samples are synthetically generated, not a copy of any samples of the minority class. this prevented the over-fitting problem. SMOTE is operative since the new artificial instances from the minority class are created, and they are quite similar in feature space to existing minority class examples. One limitation of SMOTE is the oversimplification of the minority class without seeing the majority class which may raise the intersecting between classes. There are many variants of SMOTE, which work based on it. Borderline-SMOTE [16] is a variant of SMOTE that works by only creating synthetic samples on the dividing line between two classes. Samples distant from the borderline do not help with the classification task. So, variations of borderline SMOTE are generated. Two variations of marginal SMOTE are marginal SMOTE-1 and marginal SMOTE-2. Safe SMOTE [4] works by giving a harmless label value to the positive samples earlier generating the artificial instances. SDSMOTE [24], this spatial distribution based SMOTE works mainly by generating a novel data set by acquiring raw spatial data distribution. Reverse-SMOTE based on SMOTE as well as Reverse-Nearest Neighbor (R-NN). SMOTE-Out, SMOTE-Cosine, and Selected-SMOTE are three other improved variants of SMOTE. ADASYN [18], exploits a logical method to adaptively generate different numbers of artificial instances in line with their distributions. Another way of solving the class imbalance problem is using SMOTE algorithm and applying a post-processing strategy for the modification of datasets. For example: SMOTE+Tomek, SMOTE+FRST [33].

3.2 Algorithmic Level Techniques

These types of techniques work by modifying the algorithm. For this, one needs good knowledge about the modified algorithm.
Cost-sensitive learning and one class learning are two branches of algorithmic level techniques. In the class imbalance problem, the distribution of data is not the only imbalanced but the misclassification error of an algorithm is also a crucial problem. Cost-sensitive learning can solve this problem by setting appropriate weights to the misclassification cost of the majority and the minority classes. Cost sensitive learning sets a higher cost for the misclassification cost to the minority class and creates the most cost-effective model [14]. But the misclassification error cost is often unspecified, and cost sensitive learning can lead to overfitting.
A method of cost-sensitive learning is training data modification, this works by modifying the training data by assigning weights to samples or by adjusting the decision threshold [43]. Another method of cost sensitive learning is to change the process of learning to construct a classifier that is cost sensitive. Changing the objective function of SVM with weighting tactic [8]. Incorporating a cost element into the fuzzy rule based classification system [25]. Cost-sensitive error function on the neural network [28]. There is also Cost sensitive boosting method. There is another method that is established on Bayes decision theory that works by including cost matrix with Bayes based decision boundary [9].
One class learning is another algorithmic way to solve the class imbalance problem. This is a recognition-based method. this works by training the model with only normal data, once the training is done it can find out that new examples as either normal or not normal (outliers or anomalies). Needs specialized methods to use in complex problems. One class learning is very useful when used in imbalanced data and highly dimensional noisy feature space [34].

3.3 Hybrid Techniques

This is a combination of two or more techniques working together for a better accurate model. As standard ensemble method cannot directly work for class imbalance problem.
Cluster based hybrid techniques. These techniques mainly work by creating a different cluster of the given dataset, in Cluster-based undersampling [42] Yen et al. suggested acquiring the most relevant subset of data. Other cluster-based algorithms are ClusterOSS [3] where Barrela et al. combined OSS with k-means clustering. Fast-CBUS [27] where ofek et al. proposed to cluster the minority set and create K-clusters by accruing identical majority samples as the minority samples in every cluster, Fuzzy Outlier Clustering Undersampling [40] where wang et al. proposed to use Fuzzy C-Means (FCM) clustering that gives a membership function towards

the samples which specifies their importance in the clusters.
RUSBoost [35] is a combination of RUS as well as AdaBoost, which reduces training time and increases performance. RUSMultiBoost [26] works by combining RUS with the Multi-boost ensemble model. EUSBoost [22] is a synthesis of Evolutionary Undersampling (EUS) with AdaBoost. ECUBoost [41] works by combining entropy based dynamic undersampling with RF-based boosting. RHSBoost [15] is a hybridization of RUS as well as Boosting method. EEU [44] is introduced by combining evolutionary technique as well as more than one base classifiers.
SMOTEBoost [7] is a synthesis of SMOTE and ensemble boosting procedures to improve the performance of SMOTE. SMOTEBagging [17] is a synthesis of SMOTE and ensemble bagging to increase the model's accuracy. SMOTE with SVM [10] improves the classification quality of the SVM classifier. Gaussian-Based SMOTE [23] is a combination of Gaussian distribution and synthetic data generation process. It solves the issue with the combination of Gaussian probability distribution within the feature space. K-Means and SMOTE [13] is a presentation of k-means clustering and SMOTE oversampling working together. It can outperform other popular oversampling methods. MSMOTEBoost is a combination of Adaboost and MSMOTE. MSMOTEBoost achieved a higher F1 value than SMOTEBoost.
AdaC1, AdaC2 and AdaC3 [36] is a combined strategy of adaboost and costsensitive learning. from these three algorithms, AdaC2 performs better than the other two algorithms. BoostingSVM-IB is also a combination of Adaboost and Cost-sensitive support vector machine. AdaBoost.NC [38] a new learning algorithm with negative correlation combined with AdaBoost. This also falls under cost-sensitive learning where "cost" is the diversity of degree. Mutual Information based feature selection with EasyEnsemble[MIEE] is a combined strategy of AdaBoost, Bagging, and RUS(Undersampling). The paper shows that it enhances the predicting capacity of EasyEnsemble in the case of AUC and BACC. RB-Boost [12] this is combined with AdaBoost and Hybrid sampling. Performs competitively when it is compared to other cutting-edge ensemble approaches. ESMUP-DW [2] is a mixture of AdaBoost with Hybrid sampling. The paper shows that it outperforms other detectors In Polyp detection. LEAT [29] is a novel boosting ensemble mechanism to solve imbalance problem. EnSVM [1] is a model introduced by Liu et al. It unified oversampling and undersampling with ensemble of SVM. This model is effective and they showed that in their results.

4 MULTICLASS IMBALANCE

Multiclass imbalance problem is just not as established as the binary class imbalance problem. The situation is more complicated in multiclass imbalance. There can be multiple majority and minority class. suppose an imbalance dataset has three classes A, B, C. Here, class A can be the majority class concerning class B, and also can the minority class in respect to class C. So, in multiple classes, it is hard to identify the majority and minority class concerning other classes. There are many useful applications of multiclass imbalance problem, including real-world applications. Some real-world applications of multiclass imbalance are handwriting recognition, image processing, text classification, traffic sign recognition, etc.

5 OPPORTUNITIES AND FUTURE CHALLENGES

5.1 Data Pre-Processing

Data pre-processing plays a higher role in multiclass than in binary class imbalance. In multiclass, one or more classes can get overlapped in groups because of class label noise. Decision boundaries between different classes are not well-defined. So, it is necessary to understand the relationship between classes. For instance: samples may be on the borderline for some classes and simultaneously a safe example for remaining classes. Overlapping and noisy samples must be cleaned properly. Because wrongly labeled samples can rise imbalance [5].

5.2 Standardized Evaluation Metrics

The conventional procedure of applying a single assessment metric is not adequate when dealing with a class imbalance problem even though many researchers use a number of evaluation metrics to assess their algorithms. Without using a curve-based analysis, it is hard to compare the performance of different algorithms. So, society must promote as a standard the practice of using curve-based assessment methods such as PR Curve.

5.3 Adopting Popular Algorithm for Multiclass Imbalance

Adopting popular classifier algorithms in multiclass imbalance can also be an option. But for this a deeper insight is required in imbalance distribution to understand the boundaries in classifiers [19]. Ensemble techniques should also look over to solve multiclass imbalance.

5.4 A Generalized Solution

There are many algorithms presented by scholars as a solution to the imbalance problem, some algorithm shows better results than others. But no specific algorithm performs better in all benchmarks. If an algorithm does well in one or two benchmarks that performance falls in other benchmarks. Because different imbalanced dataset has different types of imbalance distribution, distinct number of features, and distinct number of classes. So, a generalized solution for the class imbalance problem is needed. Which will perform better in all types of benchmarks and all types of the imbalance problem.

5.5 Fuzziness with Semi-Supervised Learning for Class Imbalance

Detecting the underlying imbalance in unlabeled datasets is hard. The starting group of examples may be somewhat balanced, but main distributions may be remarkably imbalanced. The first set of examples may show imbalance but do not reflect the main one. The majority class can be the minority one or there is no imbalance at all. To respond to this type of question it is possible to apply Fuzziness with semi-supervised learning. [32]. Fuzziness with semi-supervised learning can also be applied to labeling the new samples based on characteristics to decrease labeling time and effort [31], [30].

5.6 Class Decomposition Strategy for Multiclass Imbalance

Class decomposition is also a strategy for multiclass imbalance, it mainly brings down multiclass to a set of binary classes so that it can be solved with existing techniques. This has some disadvantages such as loss of balanced performance in all classes. Few researchers used this strategy for solving multiclass imbalance problem but they are mostly time consuming [45]. Researchers should look into this strategy for a more efficient technique to solve multiclass imbalance.

6 CONCLUSION

In this paper, we provided a short overview of the class imbalance problem and the most recent cutting-edge solutions of this type. We discussed the multiclass problem and why it is so hard to solve compared to binary class problem. Later we talked about some opportunities and future challenges for class imbalance problem. We hope that our discussion about the fundamental basis of the class imbalance problem, the techniques of solving the problem, and the discussion about the multiclass problem will help the researchers in this domain. Though there are many works on class imbalance learning, most of them are mainly about binary class imbalance by ignoring the multiclass imbalance problem. So, future research communities should focus more on the multiclass imbalance problem.

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